and also "$ac = 54$".

$\therefore$ we split the middle term as $6x$ and $9x$

$$\begin{aligned} 2x^2 + 15x + 27 &= 2x^2 + 6x + 9x + 27 \\ &= 2x(x+3) + 9(x+3) \\ &= (x+3)(2x+9) \end{aligned}$$

Therefore, $(x+3)$ and $(2x+9)$ are the factors of $2x^2 + 15x + 27$.

**Example 3.28**

Factorise $2x^2 - 15x + 27$

***Solution***

Compare with $ax^2 + bx + c$

$a = 2,\ b = -15,\ c = 27$

product $ac = 2 \times 27 = 54$, sum $b = -15$

$\therefore$ we split the middle term as $-6x$ and $-9x$

$$\begin{aligned} 2x^2 - 15x + 27 &= 2x^2 - 6x - 9x + 27 \\ &= 2x(x-3) - 9(x-3) \\ &= (x-3)(2x-9) \end{aligned}$$

Therefore, $(x-3)$ and $(2x-9)$ are the factors of $2x^2 - 15x + 27$.

| Product of numbers $ac = 54$ | Sum of numbers $b = -15$ | Product of numbers $ac = 54$ | Sum of numbers $b = -15$ |
|---|---|---|---|
| $1 \times 54$ | 55 | $-1 \times -54$ | $-55$ |
| $2 \times 27$ | 29 | $-2 \times -27$ | $-29$ |
| $3 \times 18$ | 21 | $-3 \times -18$ | $-21$ |
| $6 \times 9$ | 15 | **$-6 \times -9$** | **$-15$** |
| The required factors are **$-6$** and **$-9$** | | | |

**Example 3.29**

Factorise $2x^2 + 15x - 27$

***Solution***

Compare with $ax^2 + bx + c$

Here, $a = 2,\ b = 15,\ c = -27$

product $ac = 2 \times -27 = -54$, sum $b = 15$

$\therefore$ we split the middle term as $18x$ and $-3x$

$$\begin{aligned} 2x^2 + 15x - 27 &= 2x^2 + 18x - 3x - 27 \\ &= 2x(x+9) - 3(x+9) \\ &= (x+9)(2x-3) \end{aligned}$$

Therefore, $(x+9)$ and $(2x-3)$ are the factors of $2x^2 + 15x - 27$.

| Product of numbers $ac = -54$ | Sum of numbers $b = 15$ | Product of numbers $ac = -54$ | Sum of numbers $b = 15$ |
|---|---|---|---|
| $-1 \times 54$ | 53 | $1 \times -54$ | $-53$ |
| $-2 \times 27$ | 25 | $2 \times -27$ | $-25$ |
| **$-3 \times 18$** | **15** | $3 \times -18$ | $-15$ |
| $-6 \times 9$ | 3 | $6 \times -9$ | $-3$ |
| The required factors are **$-3$** and **18** | | | |

**Example 3.30**

Factorise $2x^2 - 15x - 27$

***Solution***

Compare with $ax^2 + bx + c$

Here, $a = 2,\ b = -15,\ c = -27$

product $ac = 2 \times -27 = -54$, sum $b = -15$

$\therefore$ we split the middle term as $-18x$ and $3x$

| Product of numbers $ac = -54$ | Sum of numbers $b = -15$ | Product of numbers $ac = -54$ | Sum of numbers $b = -15$ |
|---|---|---|---|
| $-1 \times 54$ | 53 | $1 \times -54$ | $-53$ |
| $-2 \times 27$ | 25 | $2 \times -27$ | $-25$ |
| $-3 \times 18$ | 15 | **$3 \times -18$** | **$-15$** |
| $-6 \times 9$ | 3 | $6 \times -9$ | $-3$ |
| The required factors are **3** and **$-18$** | | | |



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